Weather Analysis System

R Programming Project – Source Code

This beginner-friendly project reads daily weather records from a CSV file, calculates summary statistics, visualizes temperature/rainfall/humidity, examines correlations, and demonstrates a basic linear-regression model.

1. Required dataset format

Create a file named **weather.csv** with these exact column names. Temperature is in Celsius, Humidity is in percent, Rainfall is in millimetres, and WindSpeed is in km/h.

```
Date,Temperature,Humidity,Rainfall,WindSpeed
2026-01-01,28,70,2.5,12
2026-01-02,29,68,0.0,10
2026-01-03,27,78,8.2,14
2026-01-04,30,65,0.0,9
2026-01-05,26,82,12.4,16
2026-01-06,28,75,1.0,11
2026-01-07,31,60,0.0,8
```

2. R source code

```
# WEATHER ANALYSIS SYSTEM - R PROGRAMMING

# 1. Install and load the required packages
# Run install.packages() only once if the packages are not installed.
install.packages("ggplot2")
install.packages("dplyr")

library(ggplot2)
library(dplyr)

# 2. Load the weather dataset
# Keep weather.csv in the same folder as this R script.
weather <- read.csv("weather.csv")

# 3. Display and inspect the dataset
print(head(weather))
str(weather)
summary(weather)

# 4. Check for missing values
print(colSums(is.na(weather)))

# Remove rows containing missing values in the columns used below
weather <- na.omit(weather)

# Convert Date into Date format
weather$Date <- as.Date(weather$Date, format = "%Y-%m-%d")

# 5. Calculate basic weather statistics
cat("Average temperature:",
    mean(weather$Temperature), "Celsius\\n")

cat("Highest temperature:",
    max(weather$Temperature), "Celsius\\n")

cat("Lowest temperature:",
    min(weather$Temperature), "Celsius\\n")

cat("Average humidity:",
    mean(weather$Humidity), "%\\n")

cat("Total rainfall:",
    sum(weather$Rainfall), "mm\\n")

# 6. Plot temperature trends
ggplot(weather, aes(x = Date, y = Temperature)) +
  geom_line(color = "red", linewidth = 1) +
  geom_point(color = "blue") +
  labs(title = "Daily Temperature Trend",
       x = "Date", y = "Temperature (Celsius)") +
  theme_minimal()

# 7. Plot rainfall
ggplot(weather, aes(x = Date, y = Rainfall)) +
  geom_col(fill = "skyblue") +
```

```
  labs(title = "Daily Rainfall Analysis",
       x = "Date", y = "Rainfall (mm)") +
  theme_minimal()

# 8. Compare temperature and humidity
ggplot(weather, aes(x = Temperature, y = Humidity)) +
  geom_point(color = "darkgreen", size = 2) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(title = "Temperature vs Humidity",
       x = "Temperature (Celsius)", y = "Humidity (%)") +
  theme_minimal()

# 9. Calculate correlation between weather variables
correlation <- cor(
  weather[, c("Temperature", "Humidity",
              "Rainfall", "WindSpeed")],
  use = "complete.obs"
)
print(correlation)

# 10. Simple temperature prediction using linear regression
# This demonstrates a basic statistical model; it is not
# a professional weather forecast.
model <- lm(Temperature ~ Humidity + WindSpeed + Rainfall,
            data = weather)
print(summary(model))

# Predict temperature using the weather data
weather$PredictedTemperature <- predict(model, newdata = weather)

# Compare actual and predicted temperatures
ggplot(weather, aes(x = Date)) +
  geom_line(aes(y = Temperature, color = "Actual")) +
  geom_line(aes(y = PredictedTemperature, color = "Predicted")) +
  labs(title = "Actual vs Model-Predicted Temperature",
       x = "Date", y = "Temperature (Celsius)",
       color = "Measurement") +
  theme_minimal()

# 11. Save the cleaned and analyzed data
write.csv(weather, "weather_analysis_output.csv",
          row.names = FALSE)

cat("Weather analysis completed successfully!\\n")
```

3. How to run the project

1. Install R and RStudio. 2. Save the dataset as **weather.csv**. 3. Save the R code in a file such as **weather_analysis.R** in the same folder. 4. Open the script in RStudio and run it. The charts will appear in the Plots pane, and the analyzed CSV will be saved as **weather_analysis_output.csv**.

Note

The regression section is an educational demonstration based on the supplied data. It should not be presented as an accurate meteorological forecasting model. For meaningful results, use a larger, reliable weather dataset.